

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-a24c2a3feccb92096.jsonl`
- SHA-256: `c9fcea002f883cd327c87211f3a492dd18581ce015076b3c7290ffbb3ffcd36b1`
- Source modified: `2026-07-02T02:59:59+00:00`
- Imported at: `2026-07-05T16:48:24+00:00`
- project: `wf_41c8dc8a-8a4`
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:58:05.140Z)

You are an adversarial verifier in a tech-debt audit. Your default stance: REFUTE. Today is 2026-07-02.

A finder claims the following about the repo at C:/Users/avidu/Projects/khelsutra-guru/khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/:

```
{
  "title": "DEPLOY_ALPHA.md claims 'Status: LIVE' on a droplet retired 2026-06-25; GPU_VM_SERVING.md sources its secrets from that droplet",
  "category": "doc-rot",
  "severity": "P1",
  "file": "khelsutra/deploy/DEPLOY_ALPHA.md",
  "line": 3,
  "evidence": "deploy/DEPLOY_ALPHA.md:3 `***Status:** `LIVE` (deploy live-verified on the droplet 2026-06-22...)`; :35 names 'the CPU droplet 168.144.126.176' as the engine box. The droplet was retired 2026-06-25 (khelsutra commit 5ffb750 #117: 'retire the live-parity monitor ? droplet mirror decommissioned'); grep -i 'retired|decommission' over the repo's docs = 0 hits, so no doc records it. deploy/GPU_VM_SERVING.md $0 tells the operator to copy the CF-Access AUD from 'the droplet's /srv/khelsutra/config.json' and the tunnel token from 'systemctl cat cloudflared' on the droplet, and calls the droplet a 'hot fallback' ? all impossible now. Secondary rot: DEPLOY_ALPHA.md:143 claims the SPA 'shows a friendly engine offline banner (/api/health ping)' but web/src contains zero references to /api/health (grep -i health = no matches); and open issue #114 (Access sign-in broken) + #97 (split-origin breaks CF Access) contradict the runbook's shipped topology.",
  "proposed_fix": "One doc PR: flip DEPLOY_ALPHA.md status to RETIRED/HISTORICAL with a pointer to the retirement date and to issues #97/#114; rewrite GPU_VM_SERVING.md prerequisites to source the Gemini key/AUD/tunnel token from their canonical stores (Drive / CF dashboard) instead of the dead droplet; delete or soften the unimplemented /api/health-banner claim. The related api.khelsutra.guru CNAME + CF-Access application cleanup is Cloudflare-dashboard work (owner-gated) ? note it in the doc as pending.",
  "effort": "S",
  "tracked_in": "khelsutra #97, #114 (topology); droplet retirement itself untracked in-repo"
}
```

Repo root candidates: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer, C:/Users/avidu/Projects/khelsutra-guru/khelsutra, C:/Users/avidu/Projects/khelsutra-guru/sports-data-collector, C:/Users/avidu/Projects/khelsutra-guru/rally-annotator, C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport (pick from the finding's repo field; non-git dirs: 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup', C:/Users/avidu/Projects/khelsutra-guru/khelsutra-screencast).

VERIFY against the CURRENT working tree (just pulled to origin master/main): (1) does the cited code actually exist and behave as claimed at file:line? (2) was it already fixed by a recent commit (git log --oneline -15 -- <file>)? (3) is the severity honest - construct the concrete failure scenario or downgrade; (4) is the proposed fix sane and correctly scoped; (5) is it owner-gated per: alpha/serving go-live, Studio P10/P11 corpus-promotion/training,

product/strategy/licensing, real cloud spend, counsel ToS/DPA?
HARD RULES: READ-ONLY (no file modifications, no installs, no test-suite runs). gh read-only commands allowed. Your reasoning MUST cite file:line you actually read. CONFIRMED requires you personally saw the defect/debt in current code; if you cannot reproduce the claim from the code, say REFUTED; if a commit fixed it, ALREADY_FIXED.

2. user (2026-07-02T02:58:16.677Z)

DEPLOY_ALPHA.md
GPU_VM_SERVING.md
cloudflared.config.example.yml
droplet
stage_from_bucket.sh
f9eea14 web: resilient video-summary readers + show fps in the library (#140)
9223ba9 docs: refresh khelsutra docs for today's storage-track work (P2-P5) (#139)
fb2f561 web: persistent "FFmpeg not detected" banner (hard-blocker warning) (#138)
8667dbf P5: SPA storage-medium picker (store a video into your medium) (#137)
a04a21b docs(tracker): inline the D14 decisions into the P2-P5 rows (#136)

3. user (2026-07-02T02:58:16.895Z)

```
1      # Deploy KhelSutra Studio for alpha ? a shareable page, no SSH
2
3      > **Status:** `LIVE` (deploy live-verified on the droplet 2026-06-22 ?
`deploy/droplet/README.md`; only a rotated Gemini key + a browser OTP login remain to exercise
the upload?reel CUJ) . **Topology:** split ? SPA on Cloudflare Pages + engine on an `api.`
cloudflared tunnel, Cloudflare Access over both (the engine's `HOSTING_PLAN.md` $2 alpha plan).
. **Cost:** ~$0/mo standing (Pages + Tunnel + Access are free ?50 users) + Gemini per-run
(~$0.05/match).
4
5      This turns the local-only flow (run engine + SPA on your box, drive it yourself) into a
**page you can share with alpha testers** ? they open `app.khelsutra.guru`, sign in (email
allowlist), point at a video, and get a reel. No SSH, no home IP exposed, no inbound ports.
6
7      > **Outgrown the droplet?** To move the live engine onto a **GCP L4 GPU VM** (more
CPU/RAM for ffmpeg
8      > + optional WASB GPU detection) and cut the tunnel over to it, see
**[`GPU_VM_SERVING.md`](GPU_VM_SERVING.md)**
9      > ? same tunnel + Access, no DNS change, droplet kept as fallback (~$500?600/mo while
the VM runs).
10
11     ## 0. The shape
12
13     ```
14     tester browser
15     ? https
16     ?
17     Cloudflare edge (free): DNS . TLS . Access (email-OTP allowlist) in front of BOTH
hostnames
18     ??? app.khelsutra.guru ? Cloudflare Pages          (the web/ SPA; built with
VITE_API_BASE)
19     ??? api.khelsutra.guru ? cloudflared Tunnel (outbound-only) ?? 127.0.0.1:8000 on
YOUR box
20
21
22
23
24
25     - **No inbound ports.** `cloudflared` dials out; the engine stays bound to
`127.0.0.1:8000`. The box's home/public IP is never exposed.
26     - **Auth at the edge.** Cloudflare Access gates both hostnames; the engine *also*
verifies the `Cf-Access` JWT in-app (defence in depth ? `security.edge_auth_enabled`), so a
direct hit to the tunnel can't spoof identity.
```

```

27     - **Cross-origin, on purpose.** `app.` (SPA) and `api.` (engine) are different origins ?
the SPA is built with `VITE_API_BASE=https://api.khelsutra.guru/api` (PR #64) and the engine
locks CORS to the SPA origin via `RALLY_CORS_ALLOW_ORIGINS` (no engine code change ? it's
already env-driven).
28
29     > **Alternative (single-origin):** if you'd rather run **one box** that serves the SPA
*and* the API behind one origin (Caddy reverse-proxy, no Pages, no cross-origin), see
`LOCAL_APPLIANCE_ARCHITECTURE.md` D1. This runbook ships the **split** topology you chose
(stable Pages-hosted page + a thin API tunnel).
30
31     ---
32
33     ## 1. Prerequisites
34
35     - A **box** to run the engine: the CPU droplet `168.144.126.176` (Gemini-only ? no GPU;
fine for alpha) **or** your Windows+WSL GPU box (real WASB). `ffmpeg` + Python 3.8+ installed;
the `badminton-highlight-indexer` engine checked out.
36     - A **Cloudflare account** with the `khelsutra.guru` zone (you already run the marketing
site there).
37     - `cloudflared` installed on the box (`https://developers.cloudflare.com/cloudflare-
one/connections/connect-networks/downloads/`).
38     - A **fresh** `GEMINI_API_KEY` (rotate the chat-shared one ? it uploads downsized chunks
to Google; this is a paid + cloud action).
39
40     ---
41
42     ## 2. Engine (the `api.` side)
43
44     ### 2.1 Install (with the auth extra)
45     ```bash
46     cd badminton-highlight-indexer
47     pip install -e .[auth]          # PyJWT[crypto] ? REQUIRED when edge auth is on, else
every authenticated request is rejected 401
48     ```
49
50     ### 2.2 The safe-exposed config
51     Create / edit `config.json` on the box. **Where it must live:** the engine reads
`$RALLY_APP_HOME/config.json` if `RALLY_APP_HOME` is set, else `./config.json` relative to *the
directory you launch the server from* ? **not** necessarily the repo root. If the engine reads a
different (or absent) file it silently falls back to all-defaults, which means **edge auth is
OFF**. To make writer and reader agree by construction, write it with `python -m backend.main
--setup` (it targets the exact path the server reads) and edit *that* file:
52     ```jsonc
53     {
54         "security": {
55             "edge_auth_enabled": true,                // turn the CF-Access JWT verify ON
56             "allowed_video_root": "/srv/khelsutra/incoming", // the path-jail ? REQUIRED when
exposed
57             "upload_dir": "/srv/khelsutra/incoming/uploads", // MUST sit UNDER
allowed_video_root (see §7)
58             "cf_team_domain": "https://<your-team>.cloudflareaccess.com",
59             "cf_access_aud": "<the CF Access application AUD>" // filled in §4
60         },
61         "indexing": {
62             "default_segementer": "gemini" // ? CPU box MUST set this ? the shipped default
`yolo_hybrid` needs torch/GPU, and the SPA sends no segmenter, so every job fails "No module
named 'torch'" without it
63         }
64     }
65     ```
66     And export the env (a `.env` or your shell):
67     ```bash
68     export GEMINI_API_KEY="<your-rotated-key>"
69     export RALLY_CORS_ALLOW_ORIGINS="https://app.khelsutra.guru" # lock CORS to the SPA
origin

```

```

70     export RALLY_ALLOWED_HOSTS="api.khelsutra.guru,localhost,127.0.0.1" # REQUIRED ? see ?
below; without it EVERY tunneled request 400s (loopback added so on-box /api/health checks still
pass ? setting this var REPLACES the loopback-only default)
71     export RALLY_BIND_HOST="127.0.0.1" # keep the loopback bind
(cloudflared dials 127.0.0.1:8000)
72     ``
73
74     > Two env vars you MUST set, or the tunnel is dead-on-arrival. With
`edge_auth_enabled: true`, the engine (a) flips its bind to `0.0.0.0` and (b) keeps the DNS-
rebinding Host-header guard at its loopback-only default ? so it boots "healthy" but then
400`s every request `cloudflared` forwards with `Host: api.khelsutra.guru` (including the
`$6 /api/health` check). The boot only warns about this (`EXPOSED_HOST_ALLOWLIST_LOOPBACK`),
it does not stop. Set RALLY_ALLOWED_HOSTS="api.khelsutra.guru,localhost,127.0.0.1" (the
load-bearing fix ? the Host allowlist is independent of the bind; the loopback entries keep the
on-box `/api/health` check passing, since setting this var REPLACES the loopback-only default)
and RALLY_BIND_HOST="127.0.0.1" (so the bind stays loopback as $0/$6 claim; `cloudflared`
dials loopback anyway). Verified against engine `origin/master`: with the allowlist unset, `GET
/api/health` with `Host: api.khelsutra.guru` ? 400; with it set ? 200.*
75
76     > The engine refuses to start unsafe. The boot posture (engine PR #294) fails
fast** if `edge_auth_enabled` is on but the `allowed_video_root` jail or
`cf_team_domain`/`cf_access_aud` is missing, and warns if `PyJWT` isn't installed. So a
half-configured exposure can't silently come up ? fix what it prints and restart.
77
78     ### 2.3 Run
79     ``bash
80     python -m backend.main --server # binds 127.0.0.1:8000 ? kept loopback by
RALLY_BIND_HOST ($2.2); with edge_auth ON and that var unset it would bind 0.0.0.0 instead
81     ``
82     Keep it alive with a process manager (`systemd` on the droplet, mirroring
`site/scripts/provision-vm.sh`'s unit; or `pm2`/`nssm` on Windows).
83
84     ---
85
86     ## 3. The SPA on Cloudflare Pages (the `app.` side)
87
88     The SPA ships as a static Pages site, git-connected like the marketing site (auto-
deploy on merge to `master`).
89
90     1. Cloudflare dashboard ? Workers & Pages ? Create ? Pages ? Connect to Git ? pick
the `khelsutra` repo.
91     2. Build settings:
92         - Root directory: `web`
93         - Build command: `npm run build`
94         - Build output directory: `web/dist`
95         - Environment variable: `VITE_API_BASE = https://api.khelsutra.guru/api` ? this
is what points the SPA at the engine tunnel (PR #64).
96     3. Custom domain: add `app.khelsutra.guru` to the Pages project.
97
98     > Every push to `master` now rebuilds + redeploys the SPA. To preview a change first,
Pages builds a per-PR preview URL.
99
100    ---
101
102    ## 4. Cloudflare Access (gate BOTH hostnames)
103
104    1. Zero Trust ? Access ? Applications ? Add ? Self-hosted.
105    2. Application domain: add both `app.khelsutra.guru` and
`api.khelsutra.guru` to the same application (one app ? one shared session cookie, so a tester
signs in once and both the SPA and its API calls are authorized).
106    3. Policy: Action Allow, rule Emails ? your alpha testers' addresses (free ? 50
users ? fits 10?25 testers). Use one-time-PIN (email OTP) so testers need no Cloudflare account.
107    4. Copy the application `AUD` tag ? paste into the engine's `config.json`
`security.cf_access_aud` ($2.2), and set `cf_team_domain` to `https://<your-
team>.cloudflareaccess.com`. Restart the engine.

```

```

108     5. **CORS preflight ?:** the browser sends an unauthenticated `OPTIONS` preflight to
`api.` for the cross-origin POSTs. In the Access application's **Settings ? enable ?Bypass
options requests to CORS preflight?** (or add an explicit OPTIONS bypass), else the preflight is
redirected to the login page and every API call fails. The engine's CORS middleware
(`allow_credentials=false`, no cookies) then answers the preflight.
109
110     ---
111
112     ## 5. The tunnel (engine ? Cloudflare)
113
114     ```bash
115     cloudflared tunnel login                                # browser auth to your zone
116     cloudflared tunnel create khelsutra-studio              # note the <TUNNEL_UUID>
117     # copy deploy/cloudflared.config.example.yml ? /etc/cloudflared/config.yml, fill
<TUNNEL_UUID>
118     cloudflared tunnel route dns khelsutra-studio api.khelsutra.guru
119     cloudflared tunnel run khelsutra-studio                  # or install as a service:
`cloudflared service install`
120     ```
121     See `deploy/cloudflared.config.example.yml` for the ingress (only `api.khelsutra.guru ?
http://127.0.0.1:8000`).
122
123     ---
124
125     ## 6. Verify (the posture test + the CUJ)
126
127     **Exposure posture (must hold ? this is what makes it safe to share):**
128     - From OUTSIDE the box: `curl --max-time 5 http://<box-public-ip>:8000/api/health` ?
**refused / timeout** (the engine is `127.0.0.1`-bound; no inbound port). ? The box is not
directly reachable.
129     - `curl -i https://api.khelsutra.guru/api/health` **without** a CF session ? a **302 to
the Access login** (not a 200). ? The gate is in front.
130     - Open `https://app.khelsutra.guru` in a browser ? CF Access login ? after sign-in, the
SPA loads and `/api/health` returns 200 (same session cookie). ?
131
132     **The shareable CUJ (what a tester does):** open `app.khelsutra.guru` ? sign in ?
**upload a video from their device** (the SPA chunked-upload lands it in `upload_dir`, inside
the jail) ? **Process** ? watch progress ? pick rallies ? **Build** ? **download the reel**. All
from the page. *(Pasting a `gs://` URI does NOT work on this box ? see §7: the
`allowed_video_root` jail rejects every non-local URI. Browser upload is the alpha path.)*
133
134     ---
135
136     ## 7. Honest caveats / deferred
137
138     - **`upload_dir` must sit UNDER `allowed_video_root`.** If you enable browser upload, an
upload landing outside the jail is rejected by `/api/process` (a functional break, not a hole).
Keep `upload_dir` inside the jail (§2.2).
139     - **`gs://` / `s3://` / `gdrive://` URIs are REJECTED on the exposed box.** Setting
`allowed_video_root` (required to expose, §2.2) path-jails `/api/process` to *local* paths under
that root; the engine resolves a `gs://` URI to a local path that escapes the jail ? `400`
(pinned by engine `tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri`).
Cloud-URI ingest works **only** when `allowed_video_root` is unset (the local single-user
posture ? the opposite of this runbook). For alpha, the ingest path is **browser upload** (it
lands inside the jail). *(Scheme-exempting recognized buckets from the jail would be an engine
change, not a config flip.)*
140     - **You MUST set `edge_auth_enabled=true` to expose.** The boot posture (and the in-app
JWT verify) only engage then. Do not put the engine behind the tunnel with edge auth off.
141     - **Cross-origin reel download:** the reel streams from `api.` with `Content-
Disposition: attachment`, so it downloads even cross-origin ? but verify the filename in your
browser during the CUJ (cross-origin `` filename hints are ignored; the header is
the authority).
142     - **Upload size:** Cloudflare's free proxy caps request bodies at **100 MB** ? large
matches upload as ~95 MB chunks (the SPA chunked-upload path honours the engine's `max_part_mb`,
default **95**; lower `security.max_part_mb` to shrink them). `gs://` bucket pre-staging is

```

```

**not** an option on this box (see the jail caveat above).
143 - **Box offline = engine offline.** The Pages SPA stays up and shows a friendly "engine
offline" banner (`/api/health` ping); processing resumes when the box is back. Beta removes this
with a queue (`HOSTING_PLAN.md` §3).
144 - **Gemini = paid + uploads footage to Google.** Keep the key rotated; use it only for
clips you're OK sending.
145
146 ---
147
148 ## 8. Teardown
149 `cloudflared` tunnel cleanup `kheIsutra-studio` + delete the tunnel; remove the Pages
custom domain; disable the Access application; stop the engine service. The box's `output/` +
`uploads/` retain originals ? sweep them per your retention policy before any wider exposure.
150
151 ---
152 ### Changelog
153 - 2026-06-22 ? **correctness pass** (adversarial verify against engine `origin/master` +
a hermetic repro). Fixed two would-break-the-deploy gaps: (1) **`RALLY_ALLOWED_HOSTS` was
missing** ? with `edge_auth` on, the Host-header guard 400s every tunneled request until it's
set (added it + `RALLY_BIND_HOST` to §2.2; repro confirmed 400?200); (2) **`gs://` ingest is
rejected** by the `allowed_video_root` jail on the exposed box (dropped the `gs://` promise from
§6/§7; alpha path = browser upload). Minor: 401-not-500 wording (§2.1), `config.json`
location/`--setup` note (§2.2), bind comment (§2.3), ~95 MB chunk size (§7).
154 - 2026-06-21 ? created. Split-topology alpha deploy kit (cloudflared + Pages + CF
Access) consuming the engine exposure-safety boot posture (engine #294) + the SPA
`VITE_API_BASE` (#64). Pairs with `LOCAL_APPLIANCE_ARCHITECTURE.md` (single-origin alternative)
+ the engine `HOSTING_PLAN.md` §2.
155

```

4. assistant (2026-07-02T02:58:24.785Z)

DEPLOY_ALPHA.md matches the cited lines. Now let me check GPU_VM_SERVING.md, the retirement commit, and the SPA health-check claim.

5. user (2026-07-02T02:58:25.743Z)

```

1 # Move KheIsutra Studio onto an L4 GPU VM (off the droplet)
2
3 Self-serve runbook to run the live engine (behind `api./app.kheIsutra.guru`) on a **GCP
L4 GPU VM**
4 instead of the CPU droplet, and **cut the Cloudflare tunnel over** so traffic lands on
the VM. Lets
5 you switch detection between **Gemini** (cloud API, proven) and **WASB native** (uses
the L4 GPU).
6
7 This extends [`DEPLOY_ALPHA.md`](DEPLOY_ALPHA.md) (which deployed the same engine on the
droplet) ? it
8 reuses the droplet config templates in [`deploy/droplet/`](droplet/) and the GPU-VM
scripts in the
9 **engine repo** (`badminton-highlight-indexer`: `deploy/gcp/`).
10
11 > **These are manual steps today.** Automating them with a one-shot **startup-script**
you feed into
12 > the GCP console (`create_gpu_vm.sh` already injects a metadata startup-script ? today
it only installs
13 > the Ops Agent) is backlogged: engine repo `docs/BACKLOG.md` ? *Batch / declarative
orchestration*. The
14 > design hinge there is keeping the secrets (Gemini key / Access AUD / tunnel token) out
of plaintext
15 > metadata.
16
17 > **Cost ??** A `g2-standard-4` (1xL4) bills **~$0.70?0.85/hr ? ~$500?600/month** while
it runs. You
18 > keep it up, so this is an ongoing cost. **`bash deploy/gcp/teardown.sh delete` is the
only way to

```

```

19     > stop billing** (a *stopped* VM still bills its disk). For bursty/intermittent load, a
scale-to-zero
20     > **Cloud Run GPU *service*** would be far cheaper (~$0 idle) ? that's the backlogged
"M7" and a bigger
21     > build; this VM is the fast path that works today.
22     >
23     > **Cutover is safe + reversible.** Bring the VM up and **verify it serving** before
touching the
24     > tunnel; the droplet stays as a hot fallback. The tunnel is **token-managed** (ingress
lives in the
25     > Cloudflare dashboard, not on the box), so moving it is just "run the same connector on
the VM, then
26     > stop it on the droplet" ? **no DNS change, no new Cloudflare token, no CF-Access
change.**
27
28     ---
29
30     ## 0. Prerequisites (gather these once)
31
32     - **gcloud** authenticated as the project owner (`gcloud auth login`) on the machine you
run this from.
33     - **GPU quota** `GPUS_ALL_REGIONS ? 1` and the SA Ops-Agent roles ? see the engine repo
34       `deploy/gcp/README.md` §2. (You already have L4 quota=1 from the nightly runs.)
35     - **The two secrets**, which you already have from the droplet deploy:
36       - the **rotated Gemini API key** (Drive: `gemini.api_key.txt`) ? only needed for the
Gemini path;
37       - the **CF-Access AUD** + team domain ? already in the droplet's
`/srv/khelsutra/config.json`
38       (`security.cf_access_aud` / `cf_team_domain`); reuse the same values.
39       - the **tunnel token** ? in the droplet's cloudflared unit (`systemctl cat
cloudflared` ? the
40         `--token ?` value). You'll copy it to the VM in §6.
41       - **For the WASB path only:** the weights at `gs://khelsutra-rally-
corpus/models/wasb_badminton_best.pth.tar`.
42
43     You'll work in **two repo checkouts** on your machine: this one (`khelsutra`) for the
config templates,
44     and the **engine repo** (`badminton-highlight-indexer`) for the GPU-VM scripts. Commands
below say which.
45
46     ---
47
48     ## 1. Provision the L4 VM *(engine repo)*
49
50     ```bash
51     # in the badminton-highlight-indexer checkout
52     RALLY_VM_NAME=khelsutra-gpu RALLY_GPU=l4 bash deploy/gcp/create_gpu_vm.sh
53     ```
54
55     It sweeps zones (asia-south1 first ? Mumbai, near your users + co-located with the
bucket), uses a
56     driver-ready Deep-Learning-VM image (so `nvidia-smi` works immediately), and bakes in
the Ops Agent.
57     **Note the zone it prints** (e.g. `asia-south1-b`) ? you need it for SSH/teardown.
58
59     ```bash
60     ZONE=<the-zone-it-printed>
61     gcloud compute ssh khelsutra-gpu --zone=$ZONE    # confirm: nvidia-smi shows the L4
62     ```
63
64     ---
65
66     ## 2. Put the engine on the VM
67
68     The engine repo is private and the VM has no git creds, so push the source from your

```

```

machine
69      (byte-for-byte the same trick the droplet used):
70
71      ```bash
72      # in the badminton-highlight-indexer checkout
73      git archive --format=tar origin/master \
74          | gcloud compute ssh khelsutra-gpu --zone=$ZONE --command \
75          'sudo mkdir -p /opt/khelsutra/engine && sudo chown -R $USER /opt/khelsutra && tar -x
-C /opt/khelsutra/engine'
76      ```
77
78      Then on the VM (`gcloud compute ssh khelsutra-gpu --zone=$ZONE`):
79
80      ```bash
81      # ffmpeg + cv2 libs. The DLVM image is Ubuntu 22.04 ? libgl1 libgl1 libgl1 libgl1 libgl1
libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1 libgl1
82      sudo apt-get update -qq && sudo apt-get install -y ffmpeg libgl1 libgl1 libgl1 libgl1 libgl1
83      sudo mkdir -p /srv/khelsutra/incoming && sudo chown -R "$USER" /srv/khelsutra
84      cd /opt/khelsutra/engine
85      python3 -m venv /opt/khelsutra/venv
86      /opt/khelsutra/venv/bin/pip install -q --upgrade pip
87      /opt/khelsutra/venv/bin/pip install -q -e '[auth,storage-gcs]' # auth = the Cf-Access
JWT verify
88      ```
89
90      `backend.main` is torch-free, so this installs fast. (Torch only comes in with the WASB
path, §3.)
91
92      > **? Run everything as ONE user (don't reuse the droplet's `khelsutra` system user
here).** On the
93      > droplet the engine ran as a locked-down `User=khelsutra`; on this GPU VM the simplest,
mistake-proof
94      > setup is to keep the engine, the venv, **and the WASB conda env (§3) all owned by your
VM login user**
95      > (`whoami` after `gcloud compute ssh`, e.g. your Google id). A split owner is the #1
footgun here:
96      > the WASB env lands in `~/miniconda3` (your home), which a different service user can't
read, so a
97      > WASB run fails with a cryptic permission/`WASB_PYTHON not found`. §4 sets the systemd
`User=` to this
98      > same login user. (If you insist on the hardened `khelsutra` user, you must install the
venv **and**
99      > the conda env into a path it can read, e.g. under `/opt`, and `chown` them to
`khelsutra`.)
100
101      ---
102
103      ## 3. *(WASB path only)* Build the native GPU detection env
104
105      Skip this if you'll only run Gemini. For WASB-on-the-L4:
106
107      ```bash
108      # on the VM, in /opt/khelsutra/engine
109      bash deploy/digitalocean/gpu_setup.sh # builds the `wasb` conda env
(~/miniconda3/envs/wasb, torch 1.11+cu113, clones WASB-SBDT)
110      mkdir -p ~/models/WASB-SBDT/pretrained_weights
111      gcloud storage cp gs://khelsutra-rally-corpus/models/wasb_badminton_best.pth.tar \
~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
112
113      ```
114
115      The native runner needs **three** env vars (set them in `engine.env`, §4):
`WASB_PYTHON`,
116      `WASB_REPO_DIR`, `WASB_WEIGHTS`. `WASB_WEIGHTS` alone is **not** enough (validated on an
L4, 2026-06-22).
117

```



```
118    ---
119
120    ## 4. Serving config + start
```

6. user (2026-07-02T02:58:27.581Z)

```
ca8bfcc docs(deploy): consolidate GPU_VM_SERVING.md ? rebase cleanup (#119)
70083ad docs(deploy): fix $9 teardown command ? it no-ops (keeps billing) without RALLY_GPU_VM +
zone (#116)
8316798 feat(deploy): stage_from_bucket.sh ? process bucket footage on the L4 VM without a
browser upload (#115)
6e4dbaf docs(deploy): GPU_VM_SERVING.md ? move the live engine to an L4 GPU VM off the droplet
(#113)
7dc7de5 docs: freshness sweep ? reconcile 13 docs with the live repo (+ engine origin/master)
(#95)
7b04f02 deploy(kit): align with the live-verified droplet deploy (+ the gemini-default fix)
(#85)
3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook
(RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.*
tunnel + CF Access) (#65)
5ffb750 chore(ci): retire the live-parity monitor ? droplet mirror decommissioned (#117)
.github/workflows/parity.yml | 28 -----
site/scripts/parity-check.sh | 22 -----
2 files changed, 50 deletions(-)
```

7. user (2026-07-02T02:58:41.123Z)

```
5ffb750 2026-06-25 chore(ci): retire the live-parity monitor ? droplet mirror decommissioned
(#117)
```

The scheduled `live-parity` workflow compared the Cloudflare-served site (canonical, auto-deployed from git) against a second live copy on the DigitalOcean droplet `168.144.126.176:8787`. That droplet has been decommissioned (its data archived to `gs://khehsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/`), so the monitor has nothing to compare against and fails every night.

Remove the workflow and its only consumer, `parity-check.sh`. Cloudflare is now the single canonical host. `test.yml`'s ops-script check globs `scripts/\*.sh`, so dropping the script doesn't affect it.

Co-authored-by: Claude Opus 4.8 <noreply@anthropic.com>

8. user (2026-07-02T02:58:41.921Z)

```
khelsutra\web\src\App.tsx:128: // silently dropped at compile (P4 / D7). Offline (no engine) is
fine: no config ? no warning.
khelsutra\web\src\main.tsx:5:// Self-hosted OFL Noto fonts so Kannada/Hindi render crisply
offline (tracker B4 / decision D7) ?
khelsutra\web\src\api\client.test.ts:230:    vi.stubGlobal("fetch", vi.fn(async () => { throw
new Error("offline"); }) as unknown as FetchFn);
khelsutra\web\src\components\A8.status.test.tsx:102:        throw new Error("offline");
khelsutra\web\src\hooks\useIngestJob.test.ts:137:        throw new Error("offline");
khelsutra\web\src\components\build-error.test.tsx:58:        throw new Error("offline");
khelsutra\web\src\components\build-error.test.tsx:92:        if (u === "/api/query" && method ===
"POST") throw new Error("offline"); // engine down on load
khelsutra\web\src\i18n\hygiene.test.ts:146:describe("offline / self-contained ? no external
runtime assets", () => {
khelsutra\web\src\components\FfmpegBanner.test.tsx:64:        throw new Error("offline");
khelsutra\web\src\components\FfmpegBanner.tsx:9:// engine doesn't report the ffmpeg block at
all, it stays hidden ? the offline banner / wizard cover
khelsutra\web\src\components\IngestPanel.test.tsx:114:        throw new Error("offline");
khelsutra\web\src\components\QueryBar.test.tsx:4:// live parsed-pill, and the "unreadable"
```

```

branch were untested (funcs 40%). The offline parser
khelsutra\web\src\components\QueryBar.tsx:1:// The fast-entry query bar (PER_RALLY_SELECTION.md
P3). A deterministic, OFFLINE parser turns
khelsutra\web\src\components\SetupWizard.test.tsx:46: it("checks capabilities, then step 1
guides a missing Gemini key (offline-first default)", async () => {
khelsutra\web\src\components\SetupWizard.test.tsx:98: it("degrades gracefully when the engine
is unreachable (offline) ? message + a Done escape", async () => {
khelsutra\web\src\components\SetupWizard.test.tsx:103: // No step content / step indicator
when offline.
khelsutra\web\src\components\SetupWizard.tsx:4:// no key-write endpoint yet, so we tell the user
how to add the key (offline `motion` works with none).
khelsutra\web\src\components\SetupWizard.tsx:84: const [offline, setOffline] = useState(false);
khelsutra\web\src\components\SetupWizard.tsx:98: setOffline(true);
khelsutra\web\src\components\SetupWizard.tsx:116: const stepping = !loading && !offline; // the
3-step flow only when capabilities loaded
khelsutra\web\src\components\SetupWizard.tsx:142: {!loading && offline && <p
className="text-amber-700">{t("wizard.offline")}</p>}
khelsutra\web\src\components\SetupWizard.tsx:167: ) : offline ? (

```

9. user (2026-07-02T02:58:43.050Z)

No matches found

10. assistant (2026-07-02T02:59:03.208Z)

The retirement commit is confirmed (2026-06-25, droplet decommissioned, data archived). Let me check for an offline banner component, the dates of the doc commits, and the GitHub issues.

11. user (2026-07-02T02:59:07.186Z)

```

ca8bfcc 2026-06-29 docs(deploy): consolidate GPU_VM_SERVING.md ? rebase cleanup (#119)
70083ad 2026-06-23 docs(deploy): fix $9 teardown command ? it no-ops (keeps billing) without
RALLY_GPU_VM + zone (#116)
8316798 2026-06-23 feat(deploy): stage_from_bucket.sh ? process bucket footage on the L4 VM
without a browser upload (#115)
114
OPEN
Cloudflare Access: an allowlisted email still couldn't sign in (alpha auth)

```

Symptom (real demo, 2026-06-23)

Demoing the alpha (`app.khelsutra.guru`) to a friend: his email **was whitelisted** in the Cloudflare Access policy, but he **could not sign in** (the email-OTP / Access login didn't let him through), so he couldn't use the app with his own address. The owner's own email works fine.

The core flow is great ? small clip ? highlights was "pure bliss." This is pure
97

OPEN

Hosted alpha: split-origin (Pages app. + tunnel api.) breaks Cloudflare Access for the SPA ? single-origin needed

12. user (2026-07-02T02:59:07.358Z)

```

Found 3 files
khelsutra\web\src\components\FfmpegBanner.test.tsx
khelsutra\web\src\components\FfmpegBanner.tsx
khelsutra\web\src\App.tsx

```

13. user (2026-07-02T02:59:59.256Z)

Structured output provided successfully